Assumptions i) $\vec{A_0}$ at 30° of equator & $\vec{A_1}$ at 31° of equator

ii) r same (distance b/w magnet & sensor)

iii) magnetic moment ($\vec{M}$) fixed

iv) $\Delta\theta = 1^\circ$ (angle rotated by magnet)

$\Rightarrow$ field at $30^\circ = \theta_1$

$$\vec{B} = \frac{\mu_0}{4\pi}\left[\frac{3\vec{r}(\vec{M}\cdot\vec{r})}{|\vec{r}|^5} - \frac{\vec{M}}{|\vec{r}|^3}\right]$$

$$\vec{r} = \begin{bmatrix} r\cos\theta_1 \\ r\sin\theta_1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{r\sqrt{3}}{2} \\ \frac{r}{2} \\ 0 \end{bmatrix}$$

$$\vec{M} = \begin{bmatrix} 0 \\ M \\ 0 \end{bmatrix}$$

$$\therefore \vec{M}\cdot\vec{r} = \frac{Mr}{2}\cdot\cancel{\sin\theta_1} = \frac{Mr}{2}$$

$$\vec{B_1} = \frac{\mu_0}{4\pi}\left[\frac{3\vec{r}\left(\frac{Mr}{2}\right)}{|\vec{r}|^5} - \frac{\vec{M}}{|\vec{r}|^3}\right]$$

$$= \frac{\mu_0}{4\pi r^3}\left[\frac{3M\vec{r}}{2|\vec{r}|} - \vec{M}\right]$$

$$= \frac{\mu_0}{4\pi r^3}\left[3M\hat{r} - \vec{M}\right]$$

$$\vec{B_1} = \frac{\mu_0}{4\pi r^3}\left[\frac{3M}{2}\begin{bmatrix}\cos\alpha\\ \sin\alpha\\ 0\end{bmatrix} - \begin{bmatrix}0\\ M\\ 0\end{bmatrix}\right]$$

$$\vec{B_1} = \frac{\mu_0}{4\pi r^3}\left[3M\begin{bmatrix}\frac{\sqrt{3}}{4}\\ \frac{1}{4}\\ 0\end{bmatrix} - \begin{bmatrix}0\\ M\\ 0\end{bmatrix}\right]$$

$$\vec{B} = \frac{\mu_0}{4\pi r^3}\left[\frac{3\sqrt{3}}{2}M\hat{i} + \frac{3M}{2}\hat{j} - M\hat{j}\right]$$

$$\vec{B} = \frac{\mu_0 M}{4\pi r^3}\left[\frac{3}{2}\begin{bmatrix}\frac{\sqrt{3}}{2}\\ \frac{1}{2}\\ 0\end{bmatrix} - \begin{bmatrix}0\\ 1\\ 0\end{bmatrix}\right]$$

$$= \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}\frac{3\sqrt{3}}{2}\\ \frac{3}{2}\\ 0\end{bmatrix} - \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}\frac{3\sqrt{3}}{4}\\ -\frac{3}{4}\\ 0\end{bmatrix}$$

$$\vec{B_1} = \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}\frac{3\sqrt{3}}{4}\\ -\frac{3}{4}\\ 0\end{bmatrix} = \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}1.299038106\\ -0.75000\\ 0\end{bmatrix}$$

$$B_2 = \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}3\sin 31^\circ \cos 31^\circ\\ 3\sin^2 31^\circ - 1\\ 0\end{bmatrix}$$

$$= \frac{\mu_0 M}{4\pi r^3}\begin{bmatrix}1.324421389\\ -0.204207344 2\\ 0\end{bmatrix}$$

$$\cos\theta = \frac{1.87363 \cdot \cancel{x^2}}{\cancel{x^2}\sqrt{1.5}\sqrt{2.34}}$$

$$\cos\theta = \frac{1.87363}{1.4224} = 1.3215^\circ$$

Since $1.3215^\circ \neq 1^\circ$ Hence we cant use $\cos\theta$ to find angle